

NEUCHATEL, Switzerland

WMO: 066040

Lat: 47.0000N

Lon: 6.9533E

Elev: 1598

StdP: 13.87

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	23.6	12.5	11.0	24.4	16.0	13.1	27.0	23.4	39.3	21.1	38.8	6.6	70	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	85.7	67.4	82.4	66.1	79.3	64.9	68.8	81.7	67.4	79.2	66.2	76.8	5.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	96.9	73.2	63.4	92.6	72.1	62.2	88.6	71.1	33.9	82.0	32.7	79.5	31.7	77.1	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	16.7	14.4	17.5	90.4	4.0	3.6	14.6	93.0	12.2	95.1	10.0	97.1	7.1	99.8
			16.2	71.6	3.9	1.4	13.4	72.6	11.1	73.4	8.8	74.2	6.0	75.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	35.3	37.1	44.1	50.9	58.0	65.1	68.6	67.6	60.1	52.3	42.6	36.7
	DBStd	13.38	6.61	6.83	6.55	6.63	6.10	6.66	5.87	5.77	5.60	5.93	6.07	6.33
	HDD50	1786	457	362	201	69	8	0	0	0	2	45	230	412
	HDD65	5256	921	781	648	424	228	80	31	38	163	393	672	877
	CDD50	2379	0	1	18	96	257	454	576	546	306	117	8	0
	CDD65	375	0	0	0	0	12	84	142	119	18	0	0	0
	CDH74	2312	0	0	0	4	70	523	935	717	63	0	0	0
	CDH80	567	0	0	0	0	6	117	253	189	2	0	0	0
Wind	WSAvg	5.4	5.8	6.2	6.2	5.5	5.4	5.2	5.3	4.8	5.0	4.5	5.1	5.9
Precipitation	PrecAvg	38.5	3.0	2.7	2.7	2.6	3.0	4.1	3.4	4.5	3.3	2.1	4.1	3.0
	PrecMax	52.8	6.5	8.9	5.3	4.3	4.3	7.5	8.0	8.3	7.7	5.2	7.3	7.8
	PrecMin	29.2	0.9	0.4	0.3	0.7	1.4	1.3	1.3	1.3	0.7	0.1	1.3	0.1
	PrecStd	5.9	1.4	1.9	1.4	1.2	0.9	2.0	1.9	2.2	2.2	1.3	1.9	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	55.4	64.5	73.6	79.8	87.8	90.1	90.0	79.4	69.4	59.3	52.4
		MCWB	46.9	46.9	50.9	56.1	63.0	69.0	67.6	68.2	64.5	58.8	52.4	47.5
	2%	DB	49.5	51.3	60.9	69.2	75.6	83.4	86.1	84.9	75.6	65.4	55.5	49.9
		MCWB	45.3	44.5	49.1	54.3	60.8	66.3	67.3	67.2	63.6	57.6	50.6	46.4
	5%	DB	46.8	48.6	57.2	65.6	72.3	79.8	82.6	81.2	72.3	62.6	53.0	47.8
		MCWB	43.2	43.2	47.4	52.7	58.7	65.2	66.0	66.2	61.8	56.6	49.0	44.5
	10%	DB	44.3	46.0	54.0	62.1	68.8	76.5	79.2	77.7	69.1	60.2	50.7	45.5
		MCWB	41.4	41.6	46.1	51.0	57.2	63.6	64.9	64.8	60.3	55.3	47.5	42.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	47.7	52.5	58.0	64.8	70.6	70.4	70.4	66.8	60.9	53.7	48.9
		MCDB	51.0	53.7	61.5	70.4	76.6	83.8	84.5	83.5	76.1	65.8	57.1	50.9
	2%	WB	45.7	45.3	50.3	55.5	62.4	68.0	68.6	68.6	64.6	59.2	51.5	46.9
		MCDB	48.7	50.2	57.7	66.7	72.9	80.2	81.9	80.9	72.5	63.6	54.7	49.3
	5%	WB	43.6	43.7	48.7	53.7	60.2	66.3	67.3	67.2	63.1	57.6	49.7	45.0
		MCDB	46.6	48.0	55.6	63.4	69.4	77.2	79.5	78.3	70.3	61.4	52.3	47.5
	10%	WB	41.5	42.0	46.8	51.9	58.3	64.6	66.0	65.9	61.4	55.9	48.0	42.9
		MCDB	44.1	45.8	53.2	60.7	66.8	74.6	76.9	75.8	67.6	59.5	50.3	45.1
Mean Daily Temperature Range	5% DB	MDBR	6.5	9.1	12.5	14.4	14.7	15.5	16.1	15.2	13.0	9.9	7.1	6.4
		MCDBR	9.3	12.8	18.1	20.3	20.3	20.1	21.0	20.5	17.2	12.9	10.1	8.6
		MCWBR	7.0	8.0	9.4	9.6	9.7	9.0	8.6	8.4	8.3	7.4	7.2	6.7
	5% WB	MCDBR	8.9	11.3	15.7	18.5	18.2	18.8	18.8	18.1	15.3	11.3	9.3	8.1
		MCWBR	7.1	7.8	8.7	9.2	9.0	8.8	8.3	8.1	7.9	6.9	7.0	6.7
Clear-Sky Solar Irradiance	taub	0.295	0.320	0.357	0.395	0.404	0.406	0.405	0.397	0.375	0.354	0.320	0.293	
	taud	2.438	2.373	2.331	2.252	2.271	2.306	2.303	2.338	2.379	2.424	2.464	2.447	
	Ebn at Noon	254	268	273	272	273	273	271	269	264	252	241	240	
	Edh at Noon	21	28	34	40	41	40	40	37	32	27	21	19	
All-Sky Solar Radiation	RadAvg	389	657	1031	1382	1591	1884	1828	1590	1227	782	427	315	
	RadStd	48	99	154	202	148	176	160	147	112	77	55	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (105 neighbors)	N/A	N/A	N/A	+1.31	+0.72	+0.54	N/A	N/A	+95	+31

Nomenclature: See separate page